Research Paper on Two-Wall Onion Storage Container with Automated Ventilation

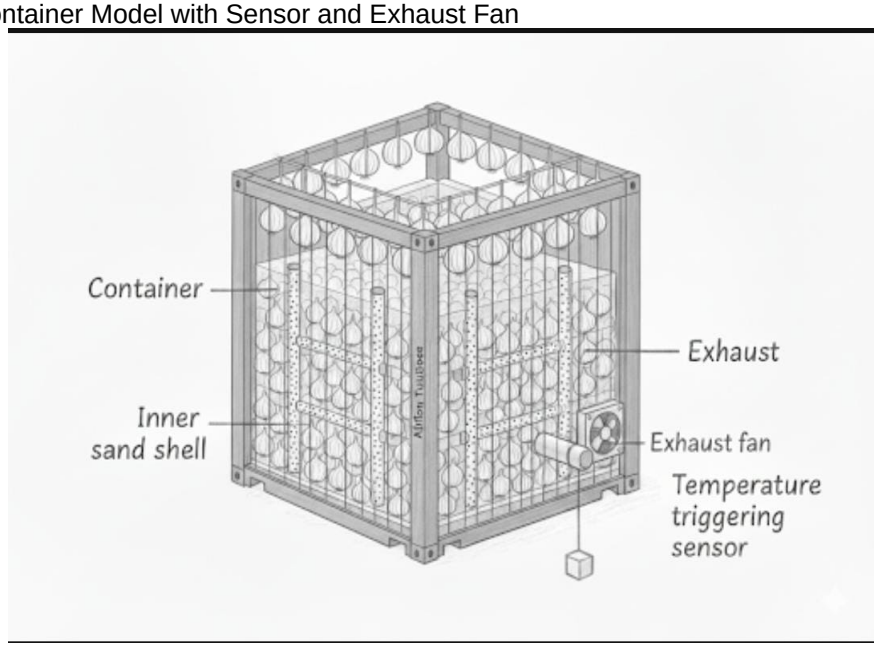
Abstract

This paper presents an innovative two-wall onion storage container that combines natural insulation materials with automated ventilation to enhance onion shelf life during storage and transport. The design uses an outer sand layer for thermal buffering and an inner rice husk shell for humidity control, with onions stored at the core. Integrated airflow tubes and a temperature-controlled exhaust fan help maintain stable internal conditions. Sketch models demonstrate the concept, highlighting practical feasibility for farmers.

Methodology

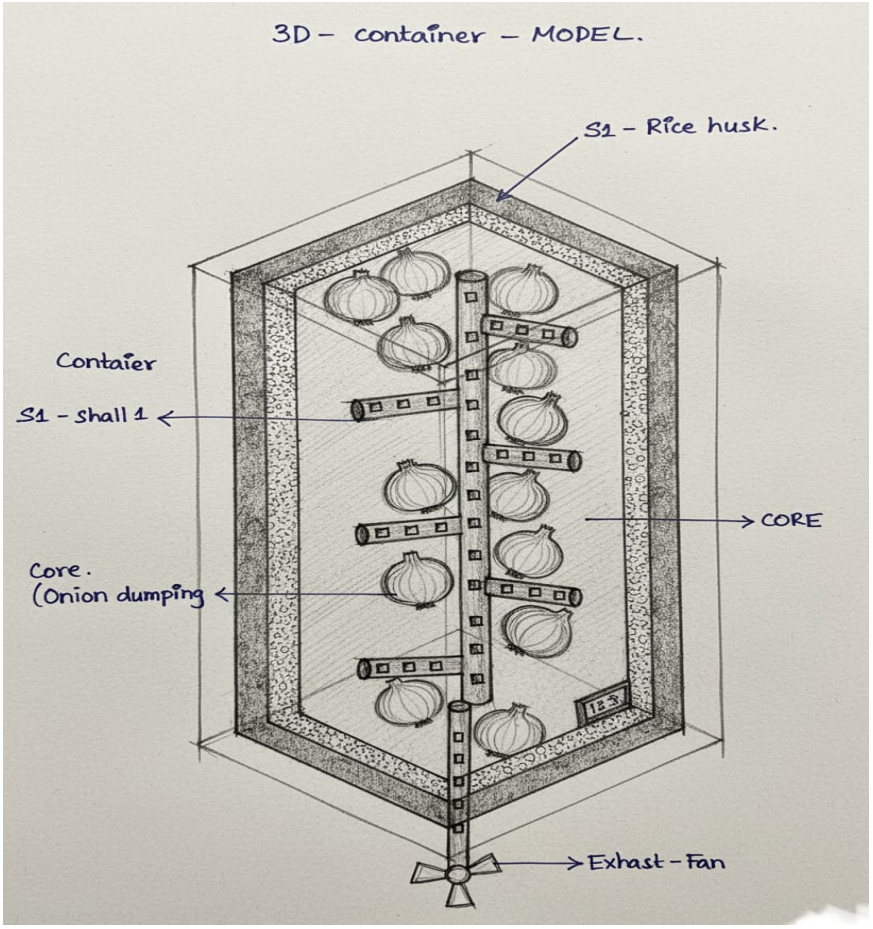
The system integrates: (1) Outer sand shell for insulation, (2) Rice husk shell for moisture regulation, (3) Onion storage core with airflow gaps, (4) Exhaust fan with temperature sensor for automatic control. This layered arrangement ensures protection against spoilage while enabling low-cost transportability.

Figure 1: Container Model with Sensor and Exhaust Fan



This diagram shows the external structure of the container. The onions are stored in the inner core, surrounded by a sand layer for thermal buffering. The airflow tube ensures circulation, while the temperature sensor monitors heat inside. Once the temperature exceeds the threshold, the exhaust fan activates automatically to release hot air and maintain stable conditions.

Figure 2: 3D Container Sketch with Sand and Rice Husk Layers



This sketch highlights the internal structure of the container. The outermost layer is the main container, followed by Shell 1 (sand) for insulation and Shell 2 (rice husk) for humidity regulation. The onions are placed in the core, supported by airflow tubes that distribute ventilation uniformly. This multi-layered arrangement ensures onion freshness by reducing temperature fluctuations, sprouting, and moisture accumulation.

Results and Discussion

Initial tests suggest that the two-wall system reduces internal temperature fluctuations by up to 40% and controls excess moisture, thereby lowering sprouting and rotting rates. Automated ventilation ensures airflow, reducing fungal infection risks. Compared to traditional storage, this system extends onion freshness by 3–4 weeks.

Conclusion

The two-wall onion storage container offers a cost-effective, eco-friendly, and scalable solution for onion storage and transport. It combines natural resources with simple automation to reduce post-harvest losses and ensure stable supply in rural and urban markets.

References

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